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THE HONG KONG
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AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 9: Transfer Function and Block Diagrams

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Transfer Function

- State Space to Transfer Function

Consider an LTI system described by state space equation

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{b}u(t)$$

$$y(t) = \mathbf{c}\mathbf{x}(t) + du(t).$$

Take the Laplace transform:

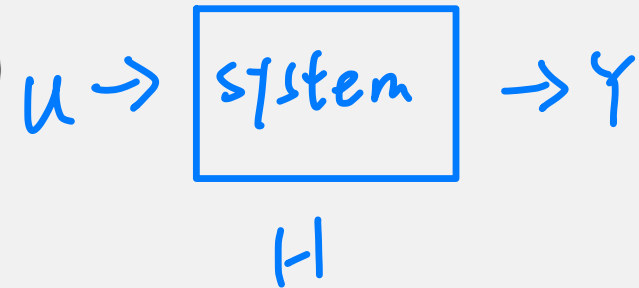
$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{b}U(s)$$

$$Y(s) = \mathbf{c}\mathbf{X}(s) + dU(s).$$

From state equation, we get

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{b}U(s).$$

A system



$\phi(t), \delta(t)$

Transfer Function

- State Space to Transfer Function

Plugging it into output equation, we obtain

$$\frac{Y(s)}{U(s)} = \mathbf{c}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{b} + d.$$

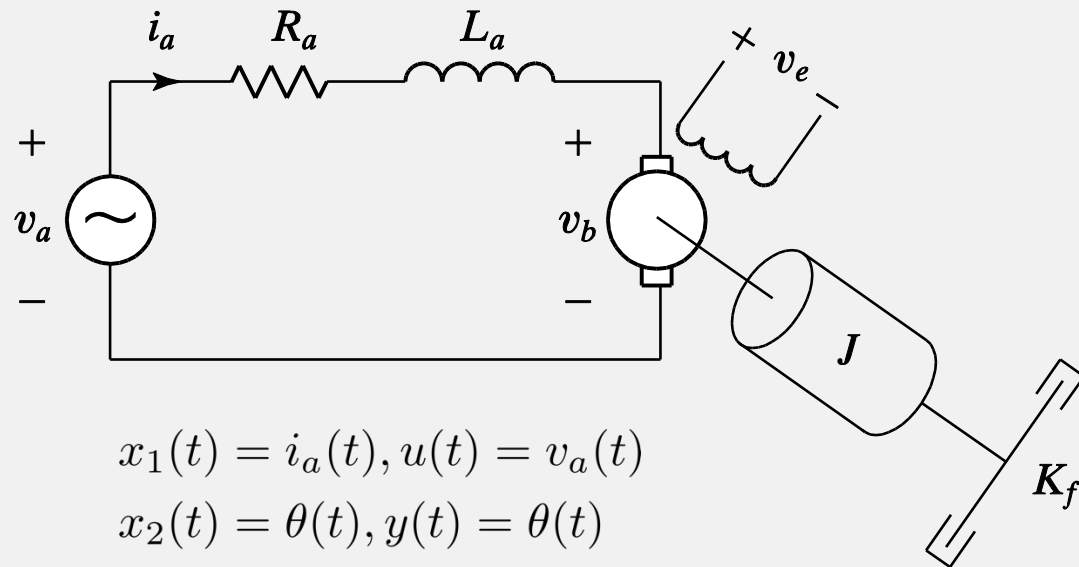
- **Definition** The **transfer function** of an LTI system is the ratio of the Laplace transform of the output over that of the input, i.e.,

$$G(s) = \frac{Y(s)}{U(s)}.$$

Transfer Function

► Example Transfer Function of the DC Motor System

- **Position Control** Recall that the state space equation is



$$x_1(t) = i_a(t), u(t) = v_a(t)$$

$$x_2(t) = \theta(t), y(t) = \theta(t)$$

$$x_3(t) = \omega(t)$$

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & 0 & -\frac{K}{L_a} \\ 0 & 0 & 1 \\ \frac{K}{J} & 0 & -\frac{K_f}{J} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

$$G(s) = C(sI - A)^{-1}b$$

Transfer Function

► Example Transfer Function of the DC Motor System

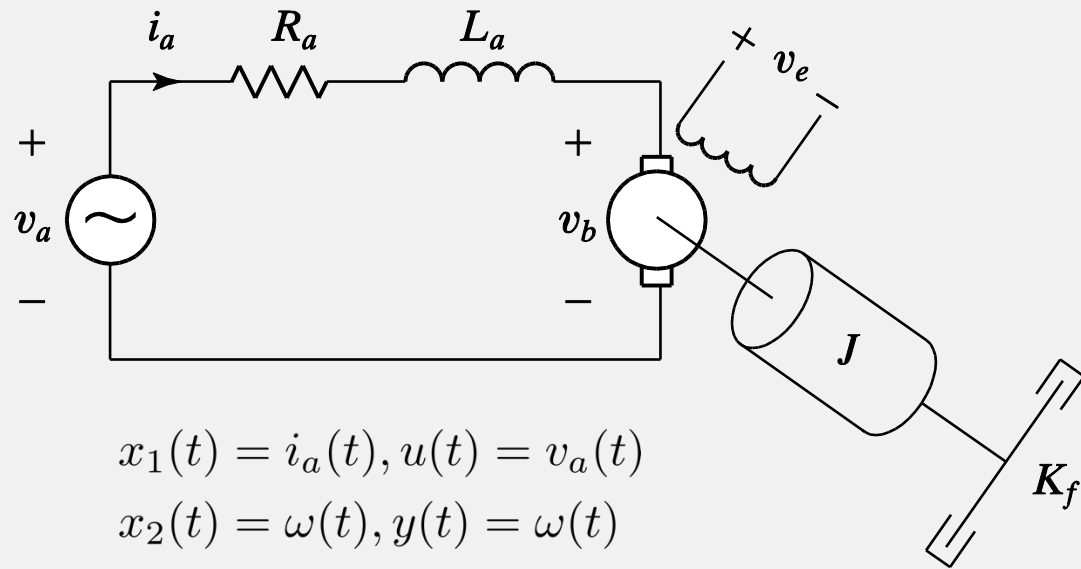
- **Position Control** The transfer function can be computed as

$$\begin{aligned} G(s) &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} s + \frac{R_a}{L_a} & 0 & \frac{K}{L_a} \\ 0 & s & -1 \\ -\frac{K}{J} & 0 & s + \frac{K_f}{J} \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{L_a} \\ 0 \\ 0 \end{bmatrix} \\ &= \frac{K/(L_a J)}{s(s + R_a/L_a)(s + K_f/J) + K^2/(L_a J)s} \\ &= \frac{K}{L_a J s^3 + (R_a J + K_f L_a) s^2 + (R_a K_f + K^2) s}. \end{aligned}$$

Transfer Function

► Example Transfer Function of the DC Motor System

- **Speed Control** Recall that the state space equation is



$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{K}{L_a} \\ \frac{K}{J} & -\frac{K_f}{J} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} \\ 0 \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

Transfer Function

► Example Transfer Function of the DC Motor System

- **Speed Control** The transfer function is given by

$$\begin{aligned} G(s) &= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} s + \frac{R_a}{L_a} & \frac{K}{L_a} \\ -\frac{K}{J} & s + \frac{K_f}{J} \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ \frac{1}{L_a} \\ 0 \end{bmatrix} \\ &= \frac{K/(L_a J)}{(s + R_a/L_a)(s + K_f/J) + K^2/(L_a J)} \\ &= \frac{K}{L_a J s^2 + (R_a J + K_f L_a) s + (R_a K_f + K^2)}. \end{aligned}$$

Transfer Function

► Example Transfer Function of MagLev

- Recall that the linearized model of MagLev is given by

$$\begin{bmatrix} \dot{\tilde{x}}_1(t) \\ \dot{\tilde{x}}_2(t) \\ \dot{\tilde{x}}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{R}{L(y_0)} & 0 & \frac{\sqrt{2rL_1Mg}}{y_0L(y_0)} \\ 0 & 0 & 1 \\ -\frac{1}{y_0}\sqrt{\frac{2rL_1g}{M}} & \frac{2g}{y_0} & 0 \end{bmatrix} \begin{bmatrix} \tilde{x}_1(t) \\ \tilde{x}_2(t) \\ \tilde{x}_3(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L(y_0)} \\ 0 \\ 0 \end{bmatrix} \tilde{u}(t)$$
$$\tilde{y} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \tilde{x}_1(t) \\ \tilde{x}_2(t) \\ \tilde{x}_3(t) \end{bmatrix}.$$

Transfer Function

► Example Transfer Function of MagLev

$$G(s) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} s + \frac{R}{L(y_0)} & 0 & -\frac{\sqrt{2rL_1Mg}}{y_0L(y_0)} \\ 0 & s & -1 \\ \frac{1}{y_0}\sqrt{\frac{2rL_1g}{M}} & -\frac{2g}{y_0} & s \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$
$$= \frac{-\frac{1}{y_0}\sqrt{\frac{2rL_1g}{M}} \frac{1}{L(y_0)}}{\left(s + \frac{R}{L(y_0)}\right) \left(s^2 - \frac{2g}{y_0}\right) + \frac{2rL_1g}{y_0^2L(y_0)}s}.$$

Rational Function

The transfer function of an LTI system with a state space model is always a ratio of two polynomials

$$G(s) = \frac{b(s)}{a(s)}$$

where $b(s)$ is called the **numerator polynomial** and $a(s)$ is called the **denominator polynomial**.

- **Coprime** $b(s)$ and $a(s)$ do **not** have common factors
- **Rational function** The ratio of two polynomials is called a **rational function**. The transfer function of an LTI system with a state space model is a rational function.

otherwise
lose some
features of
the transfer function

Transfer Function Forms

- **Zero-pole-gain Form** Denote a transfer function, or any rational function, in either of the following **unfactored form**

$$G(s) = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_m}{a_0 s^n + a_1 s^{n-1} + \dots + a_n},$$

or the **factored form (zero-pole-gain form)**

$$G(s) = K \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)}.$$

- the **zeros** of $G(s)$: $z_1, z_2, \dots, z_m$
- the **poles** of $G(s)$: $p_1, p_2, \dots, p_n$
- the **(high frequency) gain** of $G(s)$: $K(s \rightarrow \infty)$

Transfer Function

improper: $\deg(b) > \deg(a)$

- **Proper** A transfer function or system $G(s)$ is said to be **proper** if $\deg b(s) \leq \deg a(s)$, or equivalently $|G(\infty)| < \infty$.
- **Strictly proper** It is said to be **strictly proper** if $\deg b(s) < \deg a(s)$, or equivalently $G(\infty) = 0$.
- **Bi-proper** It is said to be **bi-proper** if $\deg b(s) = \deg a(s)$, or equivalently $0 \neq |G(\infty)| \neq \infty$.
- **Degree** The difference $\deg a(s) - \deg b(s)$ is called the **relative degree** of $G(s)$, and $\deg a(s)$ is called the **order** or **degree** of $G(s)$.
- **DC gain** $G(0)$ is called the **DC gain** of $G(s)$.

Transfer Function

- **Hidden poles** In the transfer function of a state space model

$$G(s) = \mathbf{c}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{b} + d = \frac{\mathbf{c} \operatorname{adj}(s\mathbf{I} - \mathbf{A}) \mathbf{b}}{\det(s\mathbf{I} - \mathbf{A})} + d$$

where adj means the adjugate of a matrix.

- If there is **no common factor** on the denominator and numerator, then $a(s) = \det(s\mathbf{I} - \mathbf{A})$, i.e., $a(s)$ is the **characteristic polynomial** of matrix $\mathbf{A}$, and the poles of the system are the eigenvalues of $\mathbf{A}$.
- Otherwise, $a(s)$ is only a factor of $\det(s\mathbf{I} - \mathbf{A})$. Hence there exists **hidden poles**.

Impulse Response

- **Impulse Response** Since the transfer function $G(s)$ of an LTI system is the ratio of the output Laplace transform $Y(s)$ and the input Laplace transform $U(s)$, i.e., $G(s) = \frac{Y(s)}{U(s)}$,

$$\text{let } U(s) = 1, \text{ i.e., } u(t) = \delta(t),$$

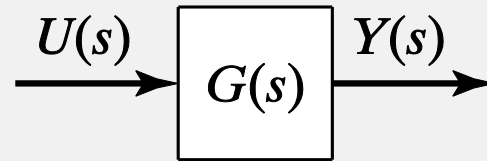
then

$$Y(s) = G(s), \text{ i.e., } y(t) = \mathcal{L}^{-1}[G(s)] = g(t).$$

Hence, $g(t)$ is called the **impulse response** of the system.

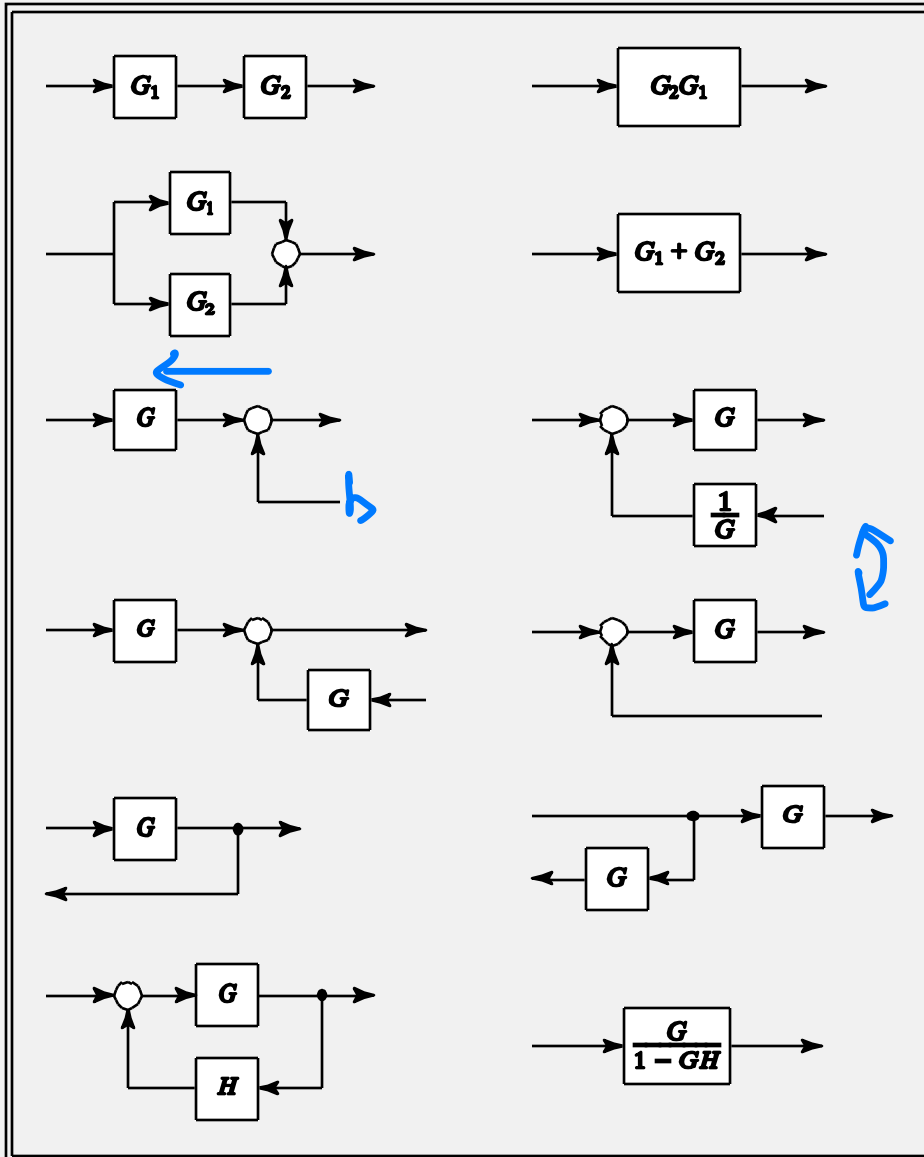
Block Diagram Representation

- Interconnected systems are often represented by **block diagrams**.
- For example, the system $Y(s) = G(s)U(s)$ can be represented by



- **Simplifying Block Diagrams**
 - Block diagrams are particularly useful when dealing with **complex** systems consisting of collections of **interconnected** subsystems.
 - They can be simplified using the **equivalence relationships**.

Equivalent Block Diagrams



口诀: junction 顺着 $\times$
node 顺着 $\div$

move junction
to left/right
same

move node to
left/right

$$G(U + YH) = Y$$

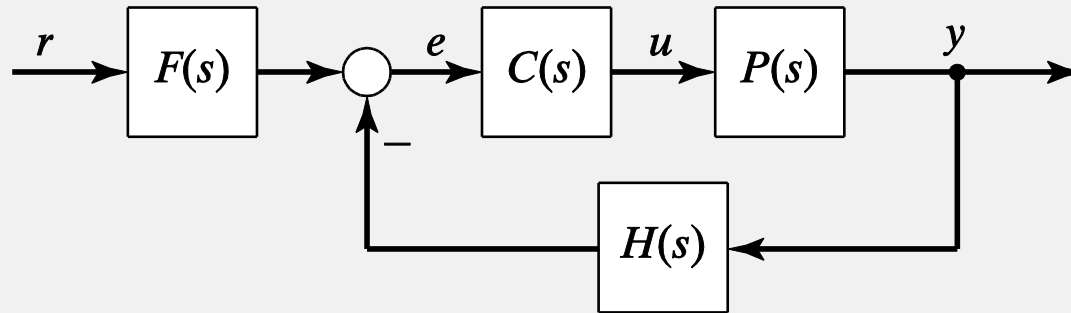
$$U + YH = \frac{Y}{G}$$

$$U = \frac{Y}{G} - YH = \frac{Y}{G}(1 - GH)$$

$$\frac{Y}{U} = \frac{G}{1 - GH}$$

Simplifying Block Diagrams

► Example A Simple Feedback System



$$\frac{Y}{R} = T = \frac{CP}{CP+1}$$

- **Equivalence relationships** To find the relationship between r and y , we shall write down all the equations:

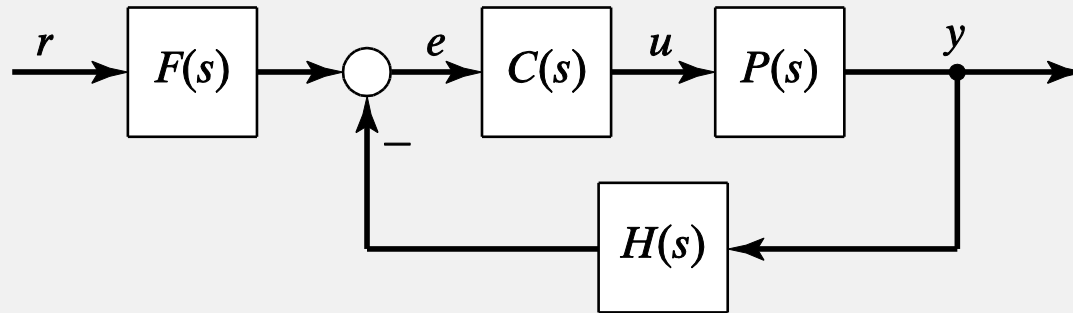
$$Y(s) = P(s)U(s), \quad U(s) = C(s)E(s), \quad E(s) = F(s)R(s) - H(s)Y(s).$$

- **Eliminate the intermediate variables** Eliminate the intermediate variables, $E(s)$ and $U(s)$, to get

$$Y(s) = P(s)C(s) [F(s)R(s) - H(s)Y(s)].$$

Simplifying Block Diagrams

► Example A Simple Feedback System



- Solve $Y(s)$

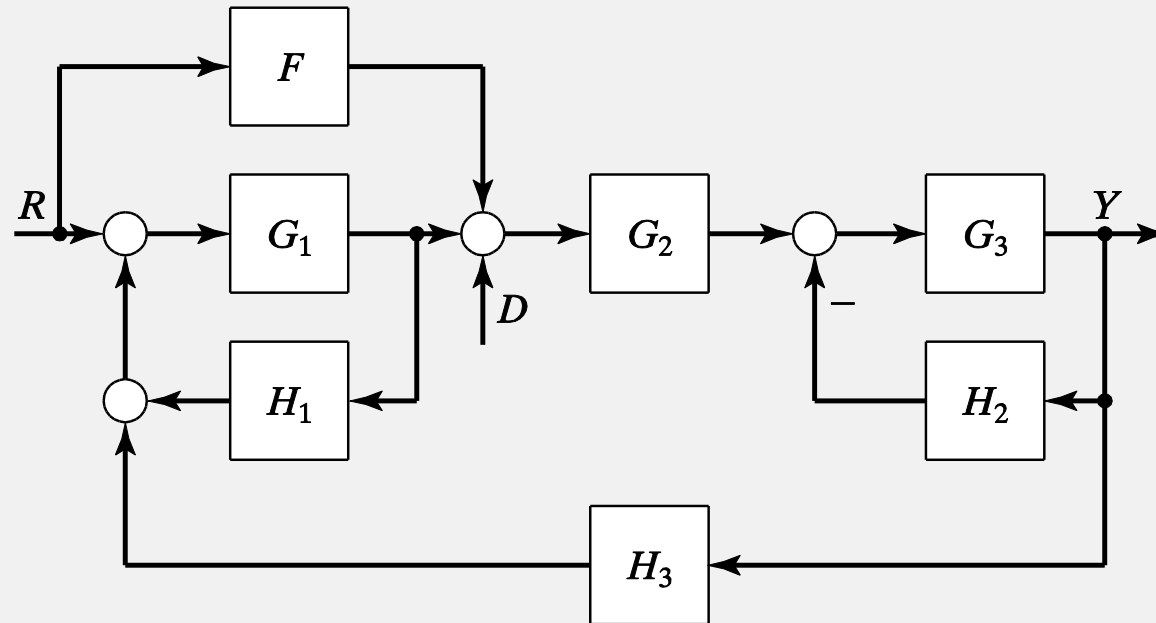
$$Y(s) = \frac{P(s)C(s)F(s)}{1 + P(s)C(s)H(s)}R(s).$$

- **Transfer function** The transfer function from $R(s)$ to $Y(s)$ is given by

$$\frac{Y(s)}{R(s)} = \frac{P(s)C(s)F(s)}{1 + P(s)C(s)H(s)}.$$

Simplifying Block Diagrams

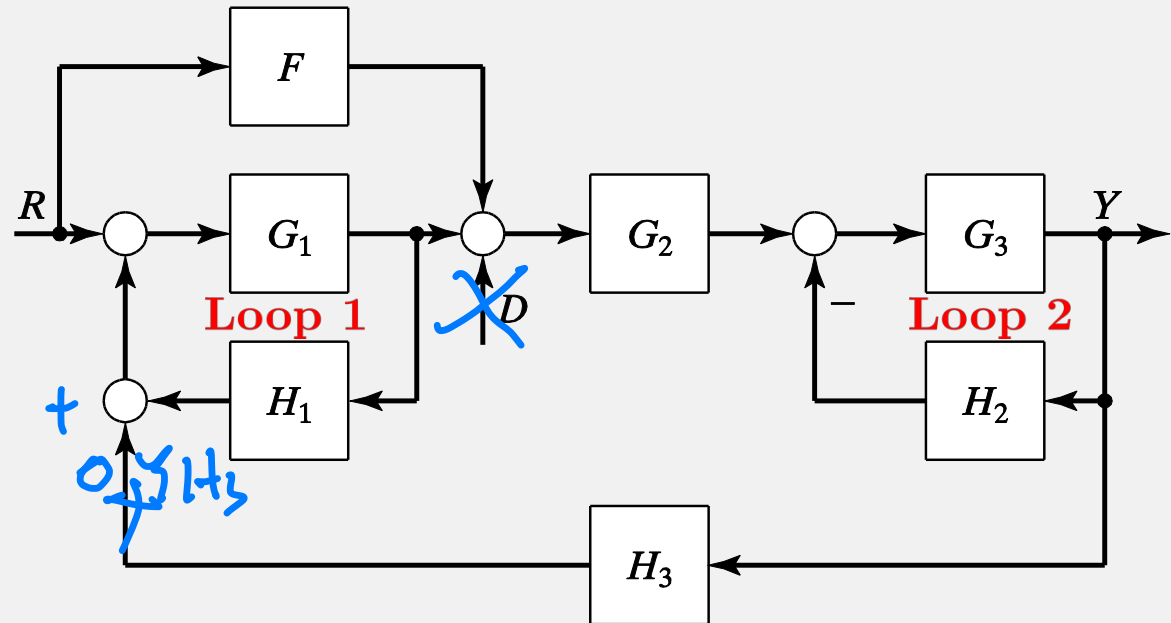
► Example A Complicated Feedback System



Compute the transfer function from R to Y and D to Y , respectively.

Simplifying Block Diagrams

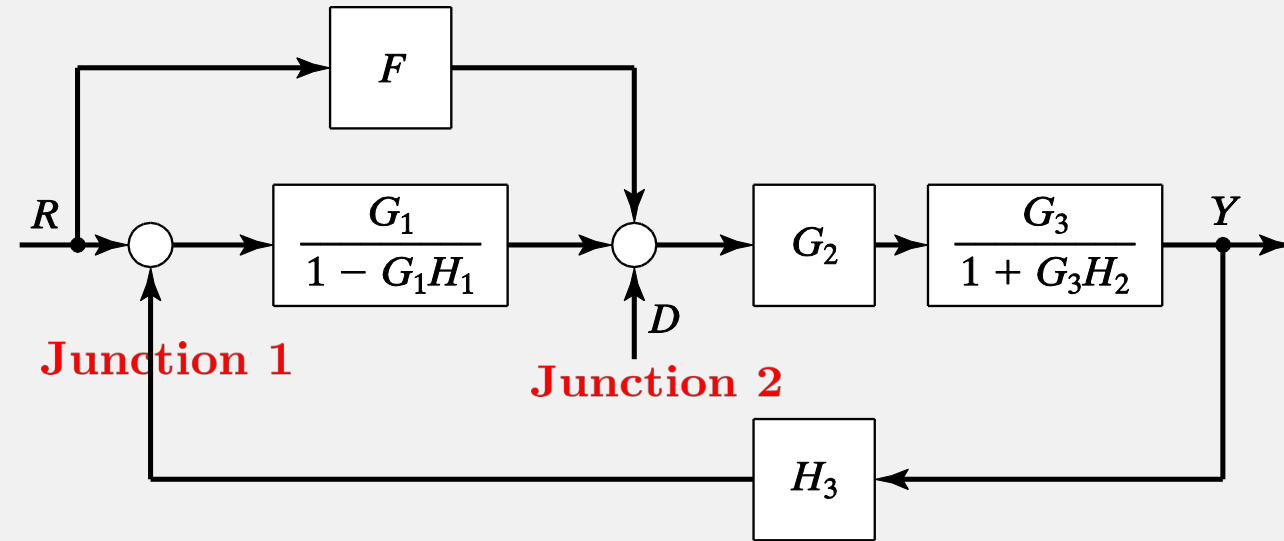
► Example A Complicated Feedback System



- Transfer function from R to Y
 - Assume $D = 0$;
 - Simplify the closing **Loop 1** $G_1 \rightarrow H_1 \rightarrow G_1$;
 - Simplify the closing **Loop 2** $G_3 \rightarrow H_2 \rightarrow G_3$;

Simplifying Block Diagrams

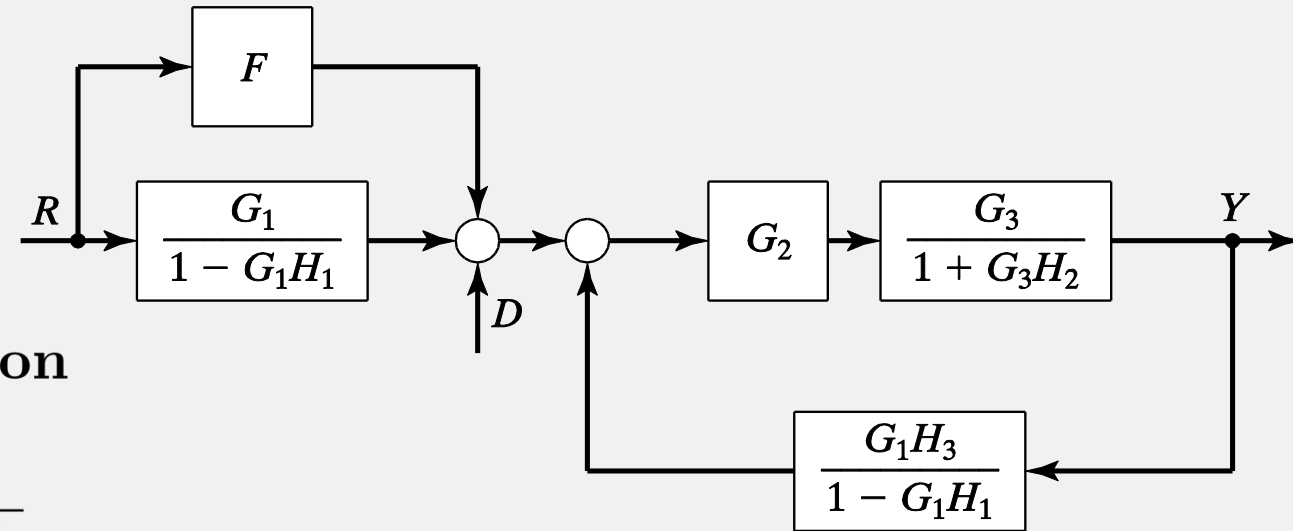
► Example A Complicated Feedback System



- Transfer function from R to Y
 - Move **Summing junction 1** to the place of **Summing junction 2**.

Simplifying Block Diagrams

► Example A Complicated Feedback System



- Find the closed-loop transfer function

$$\begin{aligned}\frac{Y}{R} &= \left(F + \frac{G_1}{1 - G_1 H_1} \right) \frac{\frac{G_2 G_3}{1 + G_3 H_2}}{1 - \frac{G_2 G_3}{1 + G_3 H_2} \frac{G_1 H_3}{1 - G_1 H_1}} \\ &= \frac{G_2 G_3 (F - F G_1 H_1 + G_1)}{1 - G_1 H_1 + G_3 H_2 - G_1 G_3 H_1 H_2 - G_1 G_2 G_3 H_3}.\end{aligned}$$

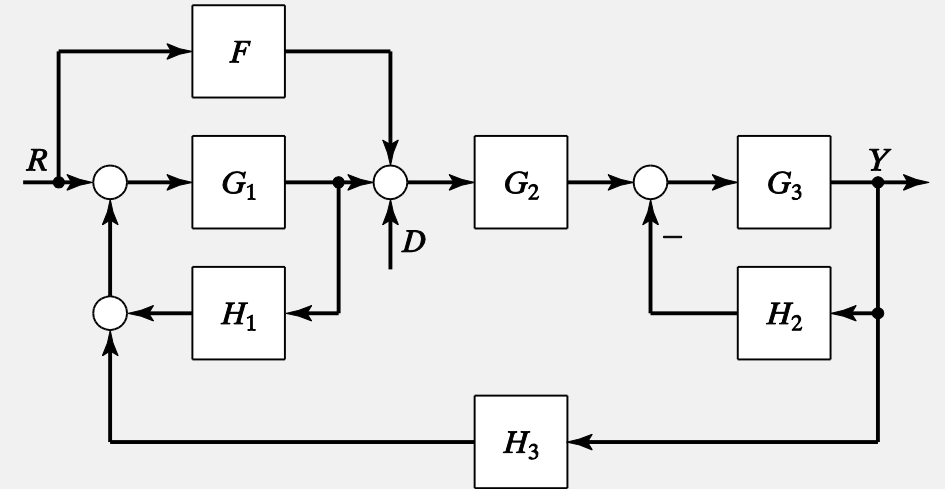
Simplifying Block Diagrams

► Example A Complicated Feedback System

- Transfer function from D to Y

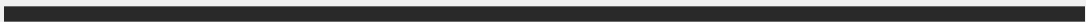
$$\begin{aligned}\frac{Y}{D} &= \frac{\frac{G_2 G_3}{1 + G_3 H_2}}{1 - \frac{G_2 G_3}{1 + G_3 H_2} \frac{G_1 H_3}{1 - G_1 H_1}} \\ &= \frac{G_2 G_3 (1 - G_1 H_1)}{1 - G_1 H_1 + G_3 H_2 - G_1 G_3 H_1 H_2 - G_1 G_2 G_3 H_3}.\end{aligned}$$

The same method!



Review & Exercise

- How to get the transfer function from the state space model?
- What's the meaning of "Coprime"?
- What's the definition of "rational function"?
- What's "zero", "pole", and "gain"?
- What's "zero-pole-gain form" or "factored form"?
- What's "Proper", "strictly proper" and "bi-proper"?
- What's "degree" and "DC gain"?
- What's "hidden poles"?
- What's the representation of "impulse response"?
- The skills of simplifying block diagrams.
- Please review the Matrix skills and do some exercises by yourself if you are not familiar with it.



Appendix

More about matrix

Inverse of a matrix

An n -by- n square matrix $\mathbf{A}$ is called invertible, if there exists an n -by- n square matrix $\mathbf{B}$ such that:

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$$

matrix $\mathbf{B}$ is called the inverse of $\mathbf{A}$, denoted by $\mathbf{A}^{-1}$.

the n -by- n identity matrix $\mathbf{I}_n =$

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

- $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
- $(k\mathbf{A})^{-1} = k^{-1}\mathbf{A}^{-1}$ for nonzero scalar k
- $(\mathbf{Ax})^+ = \mathbf{x}^+ \mathbf{A}^{-1}$ if $\mathbf{A}$ has orthonormal columns, where $^+$ denotes the [Moore–Penrose inverse](#) and $\mathbf{x}$ is a vector
- $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$
- For any invertible n -by- n matrices $\mathbf{A}$ and $\mathbf{B}$, $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$. More generally, if $\mathbf{A}_1, \dots, \mathbf{A}_k$ are invertible n -by- n matrices, then $(\mathbf{A}_1\mathbf{A}_2 \cdots \mathbf{A}_{k-1}\mathbf{A}_k)^{-1} = \mathbf{A}_k^{-1}\mathbf{A}_{k-1}^{-1} \cdots \mathbf{A}_2^{-1}\mathbf{A}_1^{-1}$.
- $\det \mathbf{A}^{-1} = (\det \mathbf{A})^{-1}$.

The adjugate of a matrix $\mathbf{A}$ can be used to find the inverse of $\mathbf{A}$ as follows: $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$

More about matrix

Determinant

The **determinant** of a square matrix **A** (denoted $\det(\mathbf{A})$ or $|\mathbf{A}|$) is a number encoding certain properties of the matrix.

$$\text{2x2} \quad \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

$$\text{3x3} \quad \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

nxn

$$\det(A) = \begin{vmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & & \vdots \\ a_{n,1} & \dots & a_{n,n} \end{vmatrix} = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1,\sigma(1)} \dots a_{n,\sigma(n)}$$

<https://en.wikipedia.org/wiki/Determinant>

Eigenvalues and eigenvectors

A number λ and a non-zero vector **v** satisfying

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

are called an *eigenvalue* and an *eigenvector* of **A**, respectively.

More about matrix

The adjugate of a square matrix $\mathbf{A}$ is the transpose of its cofactor matrix and is denoted by $\text{adj}(\mathbf{A})$.

$$\text{adj} \mathbf{A} = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix}$$

where A_{ij} is the so-called **algebraic cofactor** of a_{ij} :

$$A_{ij} = (-1)^{i+j} \det \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix}.$$

$(n-1) \times (n-1)$ matrix obtained by deleting the i th row and j th column.

Another method to calculate the inverse matrix

1. Write the augmented matrix $[A|I_n]$.
 2. Write the augmented matrix in step 1 in reduced row echelon form.
 3. If the reduced row echelon form in 2 is $[I_n|B]$, then B is the inverse of A .
- * If the left side of the row reduced echelon is not an identity matrix, the inverse does not exist.

Example: Given the matrix A below, find its inverse.

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

Solution: We write the augmented matrix as follows.

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \end{array} \right]$$

We will reduce this matrix using the **Gauss-Jordan method**. (see next slide)

Another method to calculate the inverse matrix

Solution (continue...):

Multiplying the first row by -2 and adding it to the second row, we get

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 5 & -2 & -2 & 1 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \end{array} \right]$$

If we swap the second and third rows, we get

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \\ 0 & 5 & -2 & -2 & 1 & 0 \end{array} \right]$$

Divide the second row by -2. The result is

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1/2 & 0 & 0 & -1/2 \\ 0 & 5 & -2 & -2 & 1 & 0 \end{array} \right]$$

Let us do two operations here. 1) Add the second row to first, 2) Add -5 times the second row to the third. And we get

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1/2 & 1 & 0 & -1/2 \\ 0 & 1 & -1/2 & 0 & 0 & -1/2 \\ 0 & 0 & 1/2 & -2 & 1 & 5/2 \end{array} \right]$$

Multiplying the third row by 2 results in

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1/2 & 1 & 0 & -1/2 \\ 0 & 1 & -1/2 & 0 & 0 & -1/2 \\ 0 & 0 & 1 & -4 & 2 & 5 \end{array} \right]$$

Multiply the third row by 1/2 and add it to the second. Also, multiply the third row by -1/2 and add it to the first.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -3 \\ 0 & 1 & 0 & -2 & 1 & 2 \\ 0 & 0 & 1 & -4 & 2 & 5 \end{array} \right]$$

Another method to calculate the inverse matrix

Solution (continue...):

Therefore, the inverse of matrix A is

$$A^{-1} = \begin{bmatrix} 3 & -1 & -3 \\ -2 & 1 & 2 \\ -4 & 2 & 5 \end{bmatrix}$$

Matlab to calculate the inverse matrix

`Y = inv(X)` %computes the inverse of square matrix X

Example: Compute the inverse of a matrix of symbolic numbers.

```
A = sym ([2 -1 0; -1 2 -1; 0 -1 2]);  
D = inv(A)
```



$$D = \begin{pmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{pmatrix}$$

Example: Compute the inverse of a matrix of symbolic scalar variables.

```
syms a b c d  
A = [a b; c d];  
D = inv(A)
```



$$D = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$$